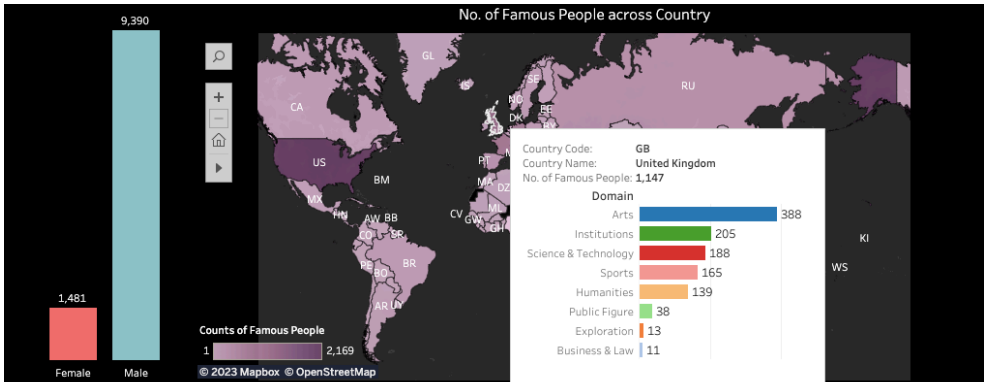
PORTFOLIO

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#DataVisualisation





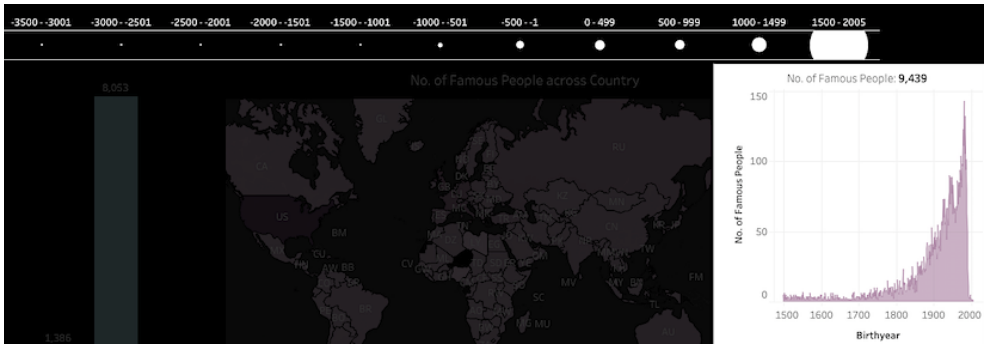


Tableau Dashboard for Pantheon /February 2022
https://public.tableau.com/views/k0726viz/Dashboard

This Tableau dashboard was built for a practical exercise in an information visualisation course. It was designed to visualise the Pantheon dataset, which is a manually verified dataset that includes 11,341 globally famous biographies.

The visualisation facilitated the exploration of the data and helped extract insights to answer the following questions:
1. *What are the demographic characteristics and occupational backgrounds of the famous biographies?*
2. *How do the relationships between HPI and page-views vary by biographies' expertise and birthplace?*

The work combines different types of charts into one dashboard and utilises a geographic map, hover-over functions, and some interactive elements such as a slider for filtering, providing more dynamic views of the data.

#DataVisualisation





Google Data (Looker) Studio Report /February 2023
https://lookerstudio.google.com/s/uJ7TFQ-KE88

This report is a replication of the performance reports previously presented to clients, using sample data provided by Google. It comprises an overview dashboard of Google Analytics data and a separate dashboard for Google Ads performance data.

The Google Analytics dashboard provides a detailed look at the acquisition performance of the account, with data on the number of users and sessions, user behaviour, and performance of pre-defined goals.

The Google Ads dashboard delivers an overview of ad performance in terms of clicks and impressions, eCommerce conversions, and overall cost. Additionally, it highlights the top performing campaigns and keywords.

#StatisticalComputing #R

Used Car Price Prediction /March 2022

https://k0726.github.io/carprices.html

This solution was built using R and R Studio as a homework assignment of a data mining course. The approach of this solution is to construct a machine learning model and evaluate its performance.

The aim of this assignment is to predict the price of used cars using vehicles listings data from Craigslist, which can be obtained from Kaggle.

In this solution, ensemble methods, Bagging and Random Forest, were chosen to model the Craigslist car dataset and forecast the response variable (price) from other given variables.

While the Random Forest model showed better results, it was selected for further enhancing the predictive performance. The enhanced model was then used to predict the price of the used vehicles.

Solution

Setup

To predict `price` from other given variables, first load relevant libraries and dataset for modelling with Random Forest.

```
# Load relevant libraries and data
library(tidyverse)
library(car) # Anova(.)
library(randomForest) #randomforest(.)
library(caret)

vehicles <- read.csv("vehicles.csv", na.strings=c("NA", "NaN", ""))
```

Before start using the dataset, some data cleaning is needed.

```
# Remove irrelevant columns and rows that with NA
dfRaw <- na.omit(select(vehicles, -url, -region_url, -VIN, -image_url, -description,
                        -county, -lat, -long, -posting_date))
# Filter out rows with extreme high / low price
dfRaw <- dfRaw %>% filter(price > 999, price < 100000)
# Set covariate "year" as factor
dfRaw$year <- as.factor(dfRaw$year)

# Use systematic sampling method to get a subset of the dataset for a faster run of model
df <- dfRaw %>% filter(row_number() %% 10 == 1)
head(df)
```

```
##           id     region price year manufacturer      model condition
## 1 7316356412     auburn 15000 2013         ford  f-150 xlt excellent
## 2 7316868220 birmingham 10950 2009        lexus      rx350 excellent
## 3 7316031977 birmingham 45000 2017    chevrolet  silverado excellent
## 4 7315259946 birmingham 98900 2001      ferrari 360 modena      good
## 5 7314833347 birmingham  4000 2002       toyota      camry  like new
## 6 7314334186 birmingham  6950 2011   volkswagen      jetta excellent
##     cylinders fuel odometer title_status transmission drive      size
## 1 6 cylinders  gas   128000        clean    automatic   rwd full-size
## 2 6 cylinders  gas   191955        clean    automatic   4wd full-size
## 3 8 cylinders  gas    92000        clean    automatic   4wd full-size
## 4 8 cylinders  gas    20187        clean    automatic   rwd  mid-size
## 5 4 cylinders  gas   160000        clean    automatic   fwd full-size
## 6 5 cylinders  gas   116000        clean    automatic   fwd full-size
##          type paint_color state
## 1       truck       black    al
## 2         SUV       white    al
## 3       truck         red    al
## 4 convertible         red    al
## 5       sedan       white    al
## 6       sedan      silver    al
```

```
# Compare the price distribution for the original dataset and its subset
par(mfrow=c(1, 2)); hist(dfRaw$price, main = "dfRaw", xlab = "Price")
hist(df$price, breaks = 20,  main = "df", xlab = "Price"); par(mfrow=c(1, 1))
```

⋮

#HTML #CSS #JavaScript

Vocabulary Game

/April 2016

http://www4.lt.cityu.edu.hk/~kmcheng49/home.html

This website features a Chinese-to-English vocabulary game and was developed as a semester-end project for a web-publishing course using the programming languages - HTML, CSS & JavaScript.



Movie Database

/November 2021

https://k0726.github.io/movies.html

This database was created for users to analyse American movies released since 1900. The program loads a local JSON dataset and includes select and search elements to filter the contents of the table.



Year	Title	Genres	Cast
1900	After Dark in Central Park		
1980	Airplane!	Comedy	Robert Hays, Julie Hagerty, Leslie Nielsen, Peter Graves, Robert Stack, Lloyd Bridges, Kareem Abdul-Jabbar
1990	The Adventures of Ford Fairlane	Comedy	Andrew Dice Clay, Wayne Newton, Priscilla Presley, Lauren Holly, Maddie Corman, Ed O'Neill, Morris Day, Tone Loc, Gilbert Gottfried, Robert Englund
1990	Sibling Rivalry	Comedy	Kirstie Alley, Scott Bakula, Bill Pullman, Jami Gertz, Sam Elliott, Carrie Fisher
1991	The Last Boy Scout	Action	Bruce Willis, Damon Wayans, Chelsea Field, Noble Willingham, Taylor Negron, Danielle Harris, Halle Berry
1991	Liebestraum	Drama	Bill Pullman, Kevin Anderson, Kim Novak
2011	The Art of Getting By	Romance, Comedy, Teen	Freddie Highmore, Emma Roberts, Michael Angarano, Elizabeth Reaser, Sam Robards, Alicia Silverstone, Rita Wilson, Blair Underwood

Card Game

/February 2022

https://k0726.github.io/cards.html

This application was developed as an assignment for an advanced programming course. It is a card game that challenges players to guess if the next card is higher or lower. The program relies on the online API to access remote services and manage the deck of cards.



#DigitalMarketing

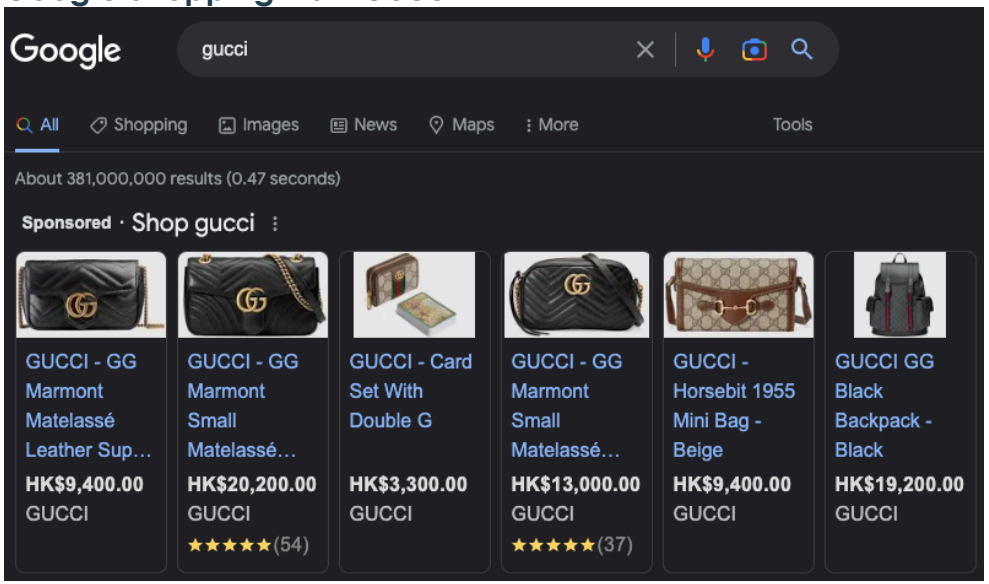
Digital Marketing Ad Examples

These are some examples of the digital marketing paid ads that I worked on as a Digital Marketing Analyst.

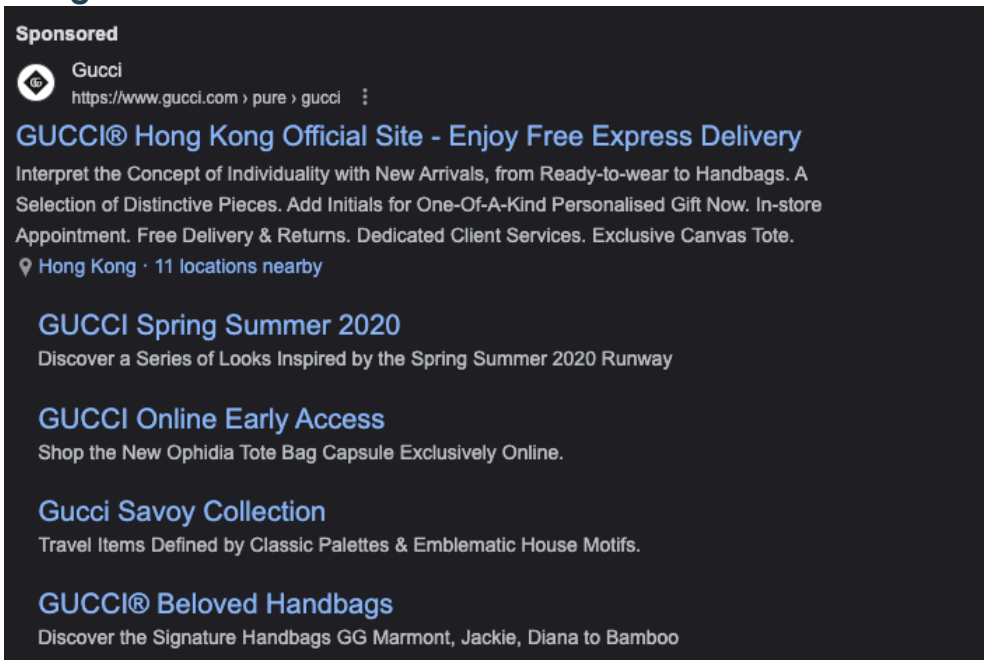
As a data-driven Digital Marketer, I have expertise in paid search (search engine marketing, SEM), paid social (social media marketing, SMM) and client management. I specialise in performance marketing, which entails analysing and optimising performance data to achieve measurable results.

I have successfully delivered outstanding results for a wide range of clients across various industries, including luxury fashion, cosmetics, eCommerce and retail.

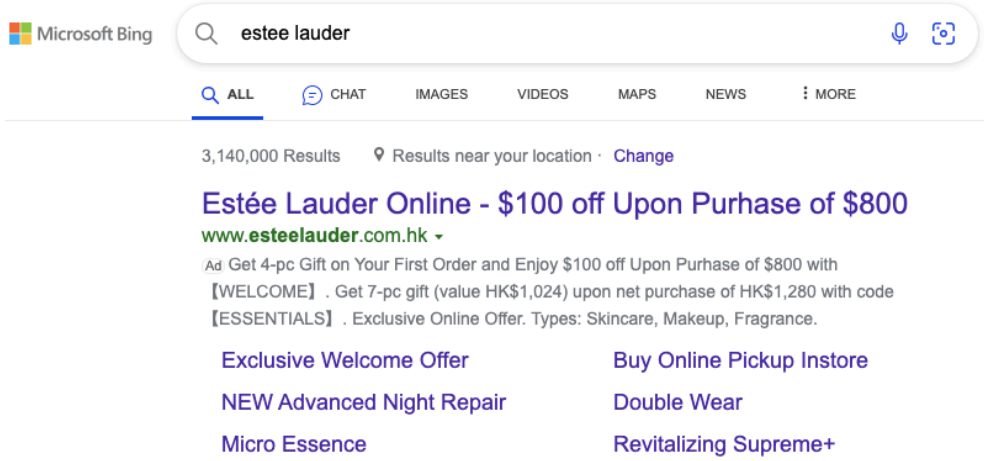
Google Shopping Ad - Gucci



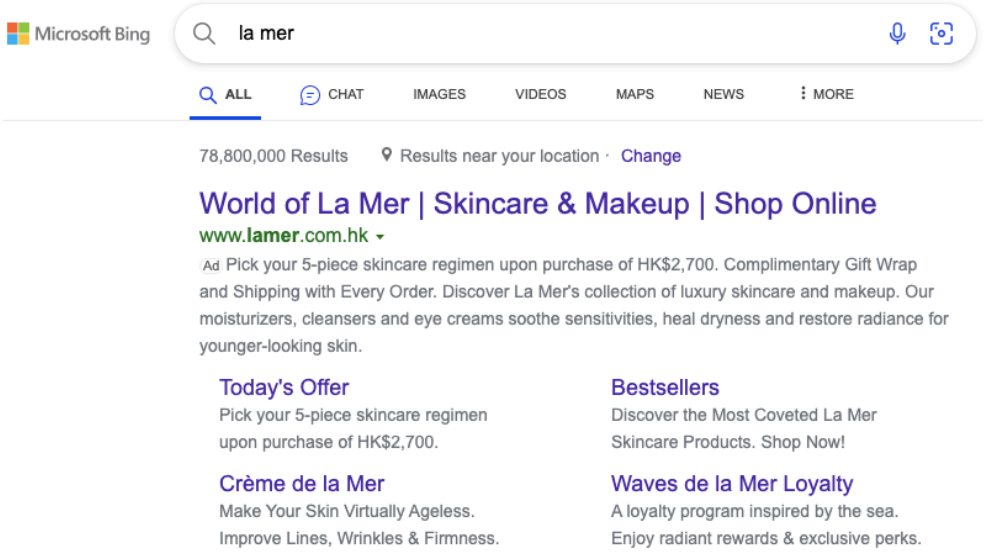
Google Search Ad - Gucci



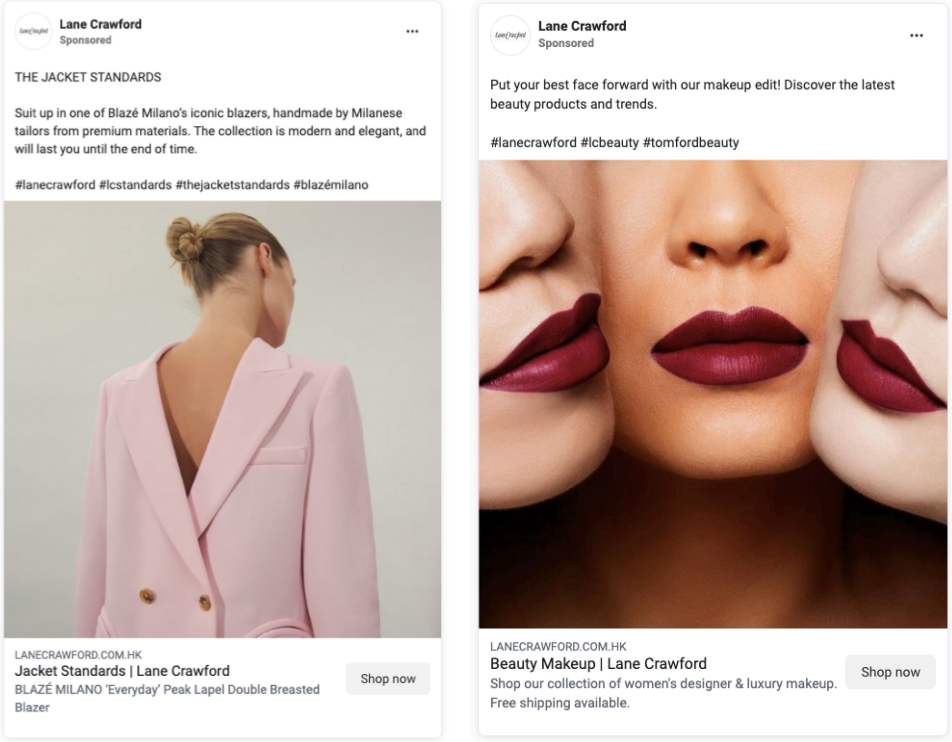
Bing Search Ad - Estée Lauder



Bing Search Ad - La Mer



Facebook & Instagram Ads - Lane Crawford



Facebook & Instagram Ads - Nike

